

CHAPTER 4

THE IMAGE PROCESSING TOOLBOX

WHAT WILL WE LEARN?

- Reading an image from file
- Main data classes used for image representation and conversion between them
- Image Display
- Explore content of an image
- Save image

The Image Processing Toolbox (IPT): essential functions and features

- Displaying information about an image file
- Reading an image file
- Data classes and data conversions
- Displaying the contents of an image
- Exploring the contents of an image
- Writing the resulting image onto a file

The Image Processing Toolbox (IPT): essential functions and features

- Displaying information about an image file

`imfinfo ('peppers.png')`

- Example 4.1 (page 62)



```
Filename: '...\peppers.png'  
FileModDate: '16-Dec-2002 06:10:58'  
FileSize: 287677  
Format: 'png'  
Width: 512  
Height: 384  
BitDepth: 24  
ColorType: 'truecolor'  
FormatSignature: [137 80 78 71 13 10 26 10]  
InterlaceType: 'none'  
Transparency: 'none'  
ImageModTime: '16 Jul 2002 16:46:41 +0000'  
Description: 'Zesty peppers'  
Copyright: 'Copyright The MathWorks, Inc.'
```


The Image Processing Toolbox (IPT): essential functions and features

- Reading an image file
 - **imread**
 - dicomread
 - nitfread
 - hdrread

MATLAB R2012b en 1:03 PM konstantin

HOME PLOTS APPS

New Script New Open Find Files Compare Import Data Save Workspace Clear Workspace New Variable Open Variable Analyze Code Run and Time Preferences Set Path Community

FILE VARIABLE

/ > home > konstantin >

```
>> I = imread('tiger.jpg');  
>> imshow(I);  
fx >>
```

Figure 1

File Edit View Insert Tools Desktop Window Help



The image shows a tiger with orange fur and black stripes lying on a dirt ground next to a light-colored tree trunk. The tiger is looking towards the camera. The image is displayed in a window titled 'Figure 1' within the MATLAB R2012b environment.

Reading and
showing an
image

Image Data Classes

- Most common data classes for images
 - uint8**: 1 byte per pixel, in the $[0, 255]$ range
 - double**: 8 bytes per pixel, usually in the $[0.0, 1.0]$ range
 - logical**: 1 byte per pixel, representing its value as true (1 or white) or false (0 or black)

Conversion between classes

- im2single
- im2double
- im2uint8
- im2uint16
- im2int16
- im2bw
- mat2gray
- ind2gray
- gray2ind
- rgb2gray
- rgb2ind
- ind2rgb

Image
conversion

vs

Type
conversion

Command Window

```
zeros(3,3)
```

```
0 0  
0 0  
0 0
```

```
, :) = ones(1,3)
```

```
0 0  
0 0  
1 1
```

s A

Size

Bytes

Class

Attribute

3x3

72

double

is an image of class double with pixel range =
uint8(A)

```
0 0  
0 0  
1 1  
im2uint8(A)
```

```
0 0  
0 0  
255 255
```

s B C

Size

Bytes

Class

Attributes

3x3

9

uint8

3x3

9

uint8

Command Window

A =

-8.0000	4.0000
0	0.5000

>> B = uint8(A)

B =

0	4
0	1

>> C = im2uint8(A)

C =

0	255
0	128

>> D = mat2gray(A)

D =

0	1.0000
0.6667	0.7083

>> E = im2bw(D,0.4)

E =

0	1
1	1

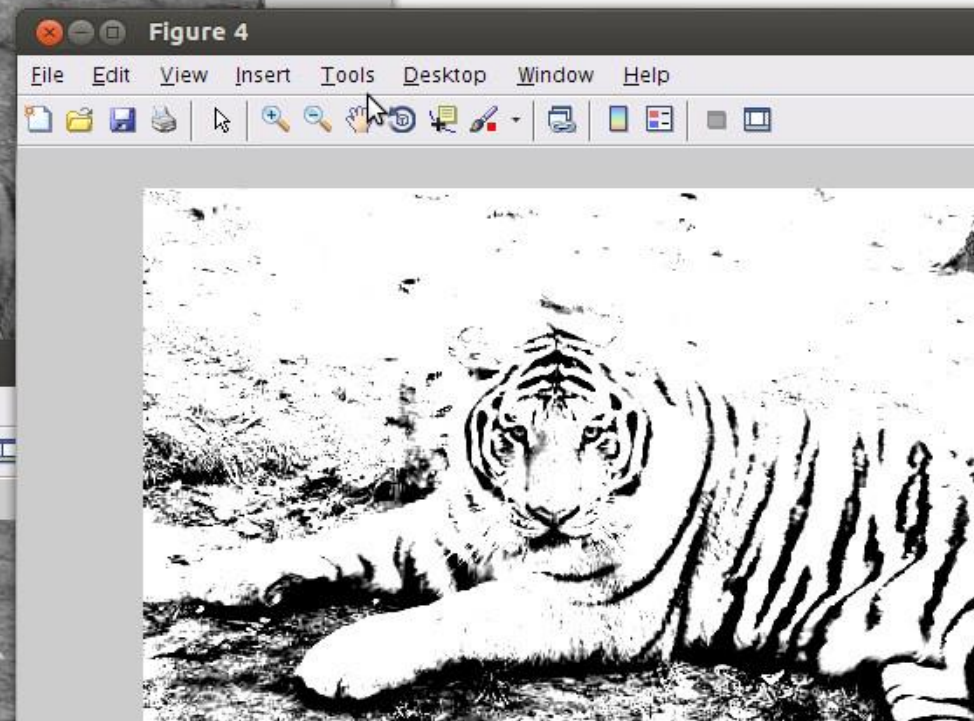
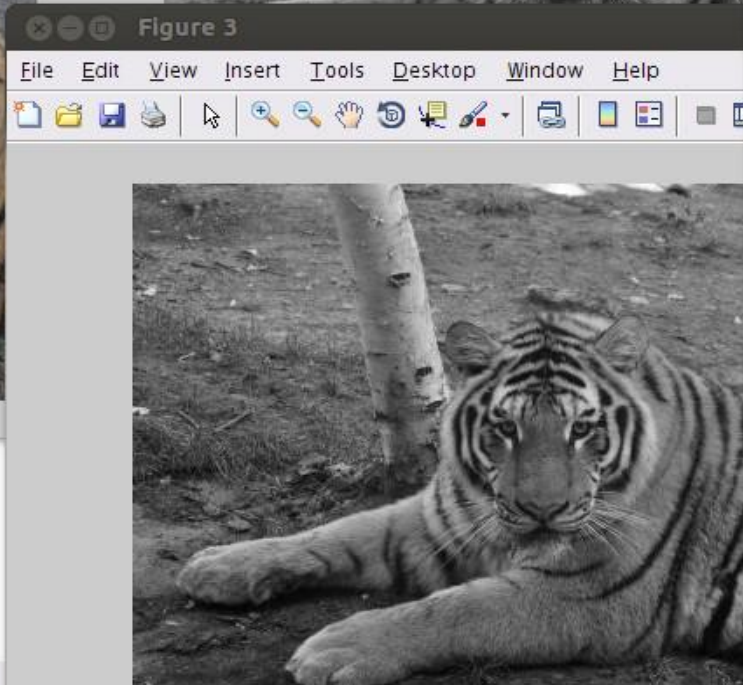
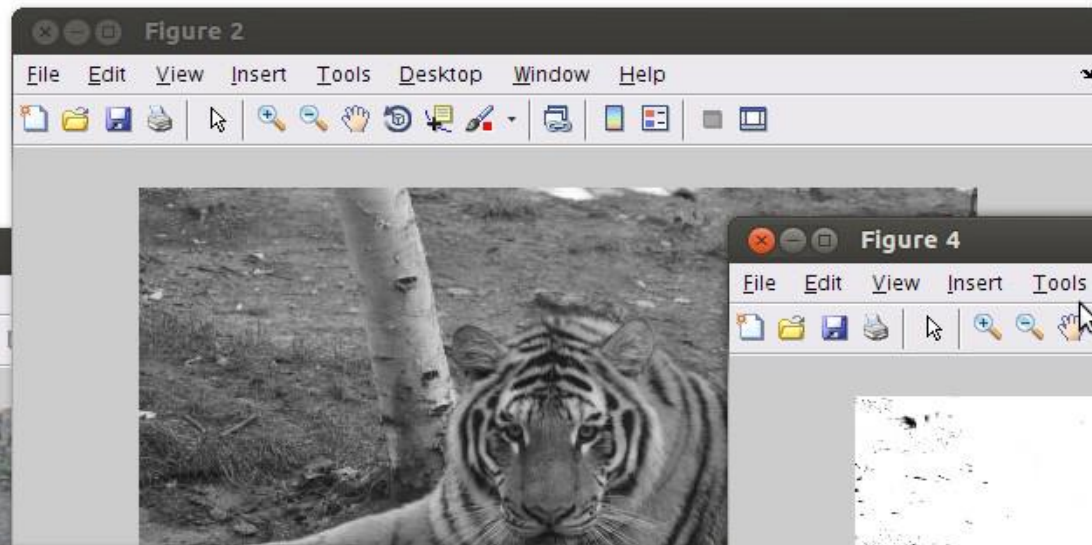
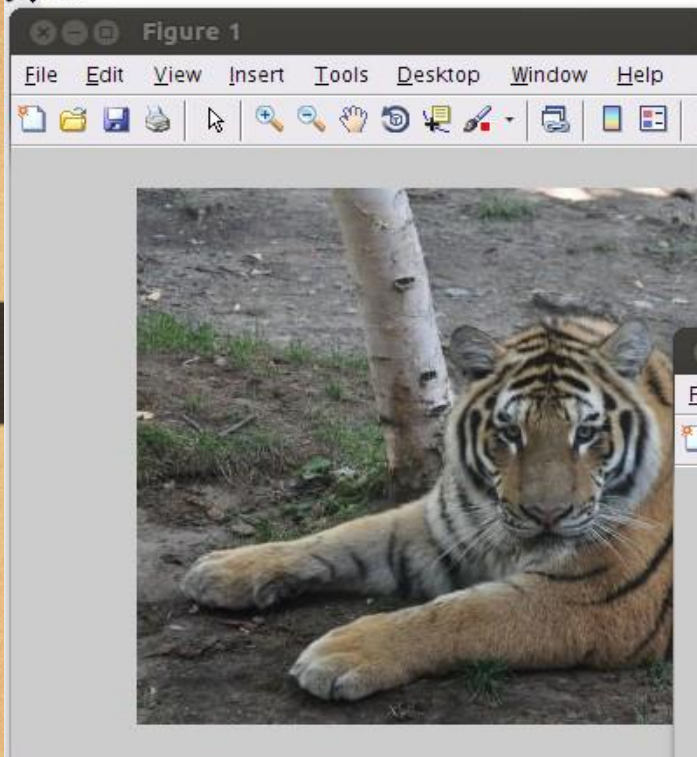
fx >>

OVR

The Image Processing Toolbox (IPT): essential functions and features

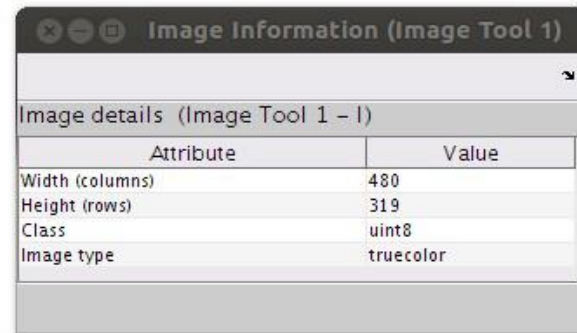
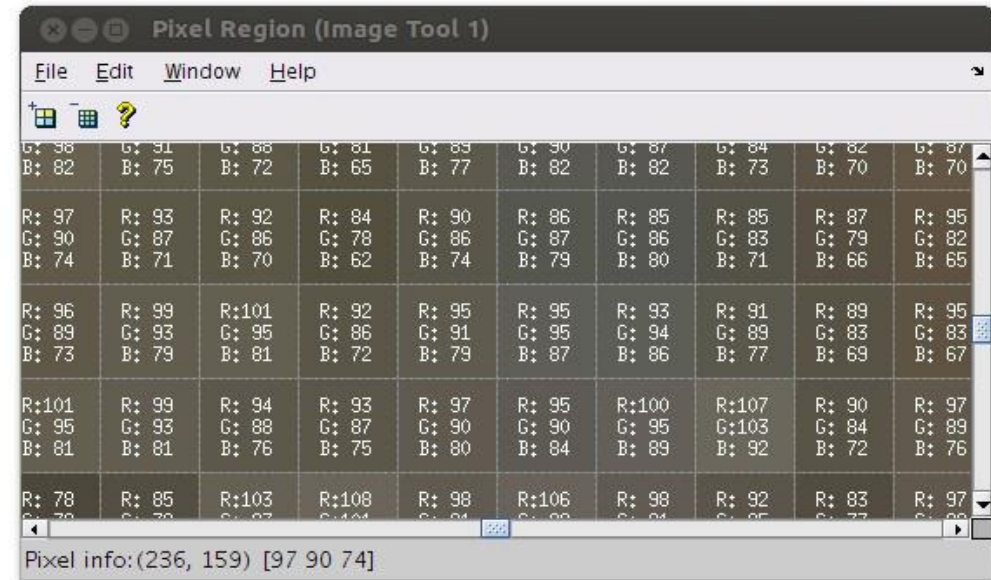
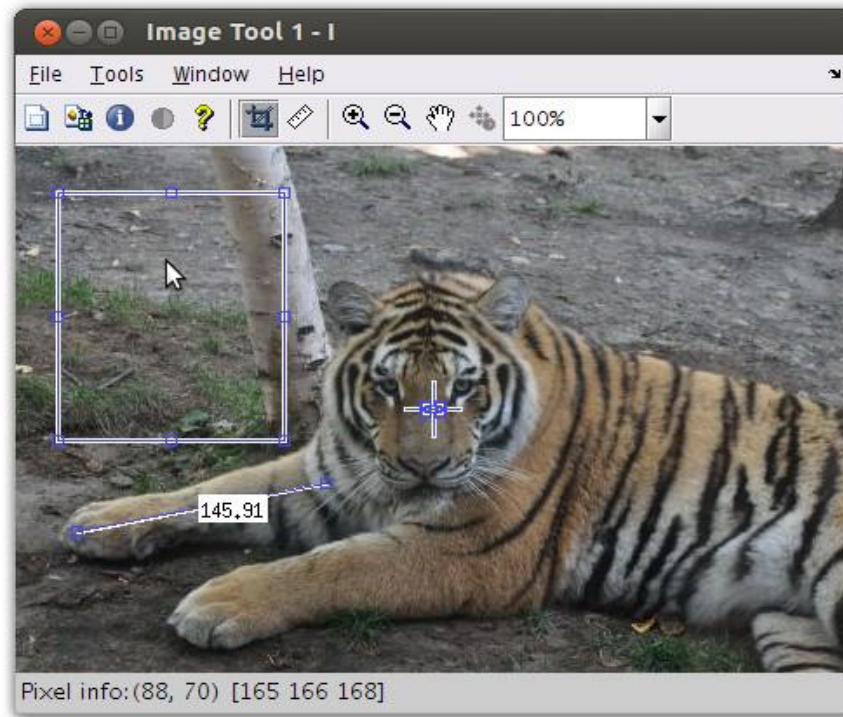
- Displaying the contents of an image
 - `image`: displays an image using the current colormap.
 - `imagesc`: scales image data to the full range of the current colormap and displays the image.
 - `imshow`: displays an image and contains a number of optimizations and optional parameters for property settings associated with image display.
 - `imtool`: displays an image and contains a number of associated tools that can be used to explore the image contents.

```
>> I = imread('tiger.jpg');  
>> Igray = rgb2gray(I);  
>> imshow(I);  
>> figure, imshow(Igray);  
>> figure, imshow(Igray,[]);  
>> figure, imshow(Igray,[50 80]);  
fx >>
```



imshow(...)


```
>> imtool(I)  
fx >>
```



imtool(...)

The Image Processing Toolbox (IPT): essential functions and features

- `imwrite`: Writing the resulting image onto a file
- Example 4.4 (page 71)

```
I = imread('peppers.png');  
imwrite(I, 'pep75.jpg');  
imwrite(I, 'pep05.jpg', 'quality', 5);  
imwrite(I, 'pep95.jpg', 'quality', 95);
```


Figures - Figure 1

File Edit View Insert Tools Debug Desktop Window Help



Figure 1 x



imwrite(...)

Command Window

```
>> P = imread('peppers.png');
>> imwrite(P, 'peps75.jpg');
>> imwrite(P, 'peps05.jpg','quality', 5);
>> imwrite(P, 'peps95.jpg','quality',95);
>> P75 = imread('peps75.png');
Error using imread (line 368)
File "peps75.png" does not exist.
```

```
>> P75 = imread('peps75.jpg');
>> P05 = imread('peps05.jpg');
>> P95 = imread('peps95.jpg');
>> figure
>> subplot(2,2,1)
>> imshow(P)
```

Warning: The initial magnification of the image is set to docked figure.

> In imshow at 239

```
>> subplot(2,2,2)
```

```
>> imshow(P75);
```

Warning: The initial magnification of the image is set to docked figure.

> In imshow at 239

```
>> subplot(2,2,3)
```

```
>> imshow(P05);
```

Warning: The initial magnification of the image is set to docked figure.

> In imshow at 239

```
>> subplot(2,2,4)
```

```
>> imshow(P95);
```

Warning: The initial magnification of the image is set to docked figure.

> In imshow at 239

```
fx>>
```

Practice and Homework

- Tutorials of **Chapter 4**
- Exercises in the end of **Chapter 4 of textbook**